BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Work Integrated Learning Programmes Division
Applied Machine Learning
(S1-26_DSEZG568)
Assignment – Problem Statement
Mobile Price Prediction- [Weightage 30%]

Instructions for Assignment Evaluation

1. Please follow the naming convention as <Group no>.ipynb.
2. Inside each jupyter notebook, you are required to mention your name, Group details and the Assignment dataset you will be working on.
3. Organize your code in separate sections for each task. Add comments to make the code readable.
4. Notebooks without output shall not be considered for evaluation.
5. Delete unnecessary error messages and long outputs.
6. Display the analysis of attributes in one frame rather than one after one. However, special treatment to attributes can be displayed separately.
7. Prepare a jupyter notebook to build, train and evaluate a Machine Learning models on the given dataset. Please read the instructions carefully.
8. Each group consists of up to 3 members. All members of the group will work on the same problem statement.
9. Only two files should be uploaded in Taxila **without zipping** them. **One is ipynb file and other one html/pdf output of the ipynb file. No other files should be uploaded.**
10. Each group should upload in Taxila to respective locations under ASSIGNMENT Tab. Assignment submitted via means other than through Taxila will not be graded.

Objective

The objective of this lab is to build and evaluate multi-class classification models to predict the price range of mobile phones based on their technical specifications.
Students will:

- Perform basic preprocessing and dataset preparation for classification
- Conduct key EDA relevant to classification
- Train multiple classification models (DT, RF, SVM, Ensemble)
- Evaluate model performance using multi-class metrics, Compare models
- Deploy in Streamlit

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Problem Statement

Dataset: Mobile Price Prediction (Train, Test)

A mobile manufacturing company wants to categorize mobile phones into four price ranges based on product specifications such as battery power, RAM, display resolution, connectivity features, etc.

Using the provided dataset, your goal is to:

a. Prepare the data for multi-class classification
b. Train and compare multiple classifiers
c. Identify the best model to predict price_range (0, 1, 2, 3)
d. Report your inferences from results and evaluation metrics
e. Deploy using Streamlit.

1. Import Libraries/Dataset [1M]

1. Download the dataset.
2. Import the necessary libraries, including numpy, pandas, sklearn, and matplotlib.
3. Load the training dataset using the pandas and display the initial few records of the dataset. Verify the target column. (Dataset provided)

2. Data and Data Models [1 M]

1. Print the shape, feature names, and data types of the dataset.
2. Print 2 rows for sanity check to identify all the features present in the dataset and if the target matches with them.

3. Data Visualization and Exploration [5 M]

i. Show suitable visualizations for dataset understanding.
ii. Comment on class imbalance using an appropriate graph.
iii. Perform statistical description of data using summary measures.
iv. Perform: Univariate Analysis (histograms, countplots)
v. Bivariate Analysis (scatterplots, boxplots), correlation heatmap
vi. Write observations describing: which features appear most influential, which variables show nonlinear patterns

4. Data Pre-processing and cleaning [5 M]

1. Do the appropriate pre-processing of the data like identifying NULL or Missing Values if any, handling of outliers if present in the dataset, skewed data etc. Encode categorical variables using: Label Encoding, One-Hot Encoding. Mention the pre-processing steps performed in the markdown cell. Explore a few latest data balancing tasks and its effect on model evaluation parameters.

2. Apply appropriate feature engineering techniques for them. Apply the feature transformation techniques like Standardization, Normalization, etc. You are free to apply the appropriate transformations depending upon the structure and the complexity of your dataset. Provide proper justification. **Techniques used without justification will not be awarded marks.** Explore a few techniques for identifying feature importance for your feature engineering task.

5. Model Building [5 M]

1. Separate features (X_train) and target labels (y_train) from the training dataset. Store the test dataset features separately as X_test. Display shapes of X_train, y_train, and X_test.the dataset into training and test sets**.**

2. Build any two appropriate models. Justify the selection of Models.

6. Performance Evaluation [2 M]

1. Do the prediction for the test data and display the results (Accuracy, Confusion Matrix, Classification Report (precision/recall/F1 for each class 0–3) for the inference. Calculate all the evaluation metrics and choose best for your model. Justify your answer. **Answers without justification will not be awarded marks.**
2. Comment on underfitting/overfitting/just right model.

7. Model Deployment [4 M]

1. Persist (save) and deploy the model you have built using Streamlit. The deployment solution should be capable of accepting requests with new feature values, querying the saved model and return the result back to the user.

Submission:

i. PDF/HTML with all outputs
ii. Source code.

8. Novelty and Innovation [2 Marks]

Two (2) marks of the total assessment are reserved for novelty and originality. These marks will be awarded to submissions that go beyond the minimum requirements of the assignment and demonstrate initiative, creativity, or technical depth. Contributions that may be considered under this criterion include, but are not limited to: well-designed UI, optimization and so on.

9. Presentation and Viva [5 M]

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